

Finn O'Shea, Ph.D.

Experience

Project Scientist

Stanford SLAC National Accelerator Laboratory

08/2021 – 08/2026

Menlo Park, CA

- Invented Cohort Organized Learning, a high-dimensional deep-learning (*pytorch*) method for unsupervised clustering and representation learning, to reduce the amount of manual feature engineering required for event-based dark matter searches.
- Helped create Coincidence Anomaly Detection (a deep-learning, *pytorch*-based method) to find edge-localized mode events without hand labels in TB-scale tokamak time-series datasets, achieving precision and recall of better than 65 percent.
- Built and instrumented (*tensorboard*) a custom scientific *python*-based particle accelerator model and *gymnasium/stable-baselines* reinforcement-learning environment to make targeting tumors with multiple accelerators possible. Presented results at LINAC 2024.
- Defined data quality protocols with accelerator device personnel to enable ML and AI application development.
- Implemented custom scientific *python*-based data cleaning on legacy time-series datasets to make ML and AI feasible: time alignment/aggregation, as well as outlier detection and removal.
- Translated student code into deployable packages, enabling real-time functionality on custom control system using scientific *python*:
 - developed code framework, including logging system to improve ease of operations;
 - managed and supported engineers using *git* ecosystem and pair-programming;
 - collaborated with UX designers to develop an easy-to-use interface for operations to take over day-to-day use;
 - containerized code using *docker*.

Senior Scientist

Nusano, Inc.

10/2019 – 08/2021

Valencia, CA

- Developed \$10M budget and schedule for construction of high-energy beam transport lines (including design, purchasing, installation).
- Implemented beam-tracking simulations for design of multi-target radioisotope production facility to reduce operational risk. Saved millions and made the company much more likely to be successful.
- Collaborated with others to build the accelerator. Liaised with component vendors, construction team, controls engineers, target systems group, and radiation safety group.
- Designed radio-frequency beam measurement devices to improve resolution and space efficiency, and maximize signal and information transmission.

Researcher

Elettra-Sincrotrone Trieste, S. C. p. A.

10/2018 – 10/2019

Trieste, Italy

- Developed from-scratch first-principles reinforcement learning code in scientific *python* to enable improved control of free-electron laser that reduced beam line optimization time from hours to minutes.
- Constrained by security concerns to scientific *python* only, no *tensorflow* or *pytorch*:
 - implemented models from literature;
 - built custom training environments for fast model evaluation;
 - built *python-to-tango* interface to control billion dollar machine.
- Trained deep neural network in *pytorch* to predict laser output energy, demonstrating unsuitability of supervised learning to real-time applications during operations which motivated the change to reinforcement learning.

Research Scientist

December 2012 – August 2018

Radiabeam Technologies

Santa Monica, CA

- Defined manufacturing tolerances for cryogenic undulators to reduce cost of terawatt-class free-electron lasers.
- Designed magnetic insert which enabled testing of integrable optics principle at IOTA ring, including managing engineering team throughout manufacturing.
- Designed cryogenic target for compact radiotherapy systems and led proton acceleration experiment at a high-power laser.
- Invented a novel acceleration method for high-brightness electron beams, defining scaling laws.
- Devised novel brightness measurement method for ultra-short (femtosecond), low-emittance (sub-micron) electron beams.
- Designed and built automated measurement system for magnetic centers of low field accelerator magnets; enabled center-finding with 10-micrometer precision on 10 G fields.

Education

Doctorate (Ph. D.) in Physics

12/2012

University of California, Los Angeles

- **Dissertation:** Compact radiation sources for increased access to high brightness X-rays.
- **Advisor:** James Rosenzweig

Masters (M. Sc.) in Physics

12/2006

University of California, Los Angeles

Bachelors (B. Sc.) in Physics

12/2004

San Diego State University

Skills

- **Subject matter expert: Accelerator physics.**
- **Codes:** Expert w. python; skilled w. pytorch, gymnasium, jupyter, pandas, numpy, matplotlib, scipy, docker, scikit-learn; familiar w. stable-baselines, git, Bokeh, Tensorflow, Keras, Tensorboard.
- **Accelerator codes:** Skilled with elegant; familiar with Opal, TraceWin, cheetah, Spiffe.
- **Engineering codes:** Skilled with matlab, octave, Mathematica; familiar with LabVIEW.
- **Coursera:** (U. Toronto) Self-Driving Cars Specialization (CV8G256FKCZQ); Machine Learning (HCEBSTZTDUTX); SQL for Data Science Credential (EAJASUFGYBM3).

Personal skills

- Training, mentoring. Positive reinforcement, constructive criticism. Managing, directing, supervising. Delegating. Decision-making. Project management.
- Problem-solving. Root cause analysis. Process improvement. Pattern recognition. Lateral thinking.
- Communication skills. Presentation, clarity, public speaking, active listening.
- Organizational skills. Documentation. Knowledge management.
- Strong work ethic. Responsible, disciplined. Self-starter. Initiative.
- Prioritizing, goal-setting, multi-tasking. Results-oriented.
- Time management. Planning, scheduling, logistics.
- Conflict resolution. Patient, tolerant, diplomatic.
- Negotiating, facilitating. Cooperating, collaborating, coordinating.
- Work well under pressure. Crisis management.

References and publications available upon request

Selected publications

- F. H. O'Shea** and M. E. Monzani, *Cohort Organized Learning: Clustering Through Agreement*, Mach. Learn.: Sci. Technol. **7**, 035057 (2026); arXiv: 2606.21743.
- J. Liang, W. Colocho, F.-J. Decker, R. Humble, B. Morris, **F. H. O'Shea**, D. A. Steele, Z. Zhang, E. Darve, D. Ratner, *Coincident learning for beam-based rf station fault identification using phase information at the SLAC linac coherent light source*, Phys. Rev. Accel. Beams **28**, 124601 (2025).
- M. Song, F. Bai, Y. Hu, **F. H. O'Shea**, D. Ratner, R. Smith, and G. Wang, *Model-driven prediction for accelerator magnet diagnostics to improve operation reliability*, Nucl. Instr. Meth. A **1082**, 171036 (2025).
- F. H. O'Shea**, S. Joung, D. R. Smith, D. Ratner, R. Coffee, *Coincidence anomaly detection for unsupervised locating of edge localized modes in the DIII-D tokamak dataset*, Mach. Learn.: Sci. Technol. **5**, 035050 (2024).
- S. Joung, D. R. Smith, G. McKee, Z. Yan, K. Gill, J. Zimmerman, B. Geiger, R. Coffee, **F. H. O'Shea**, A. Jalalvand, and E. Kolemen, *Tokamak edge localized mode onset prediction with deep neural network and pedestal turbulence*, Nucl. Fusion **64** 066038 (2024).
- R. Humble, W. Colocho, **F. O'Shea**, D. Ratner, and E. Darve, *Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source*, EPJ Web of Conferences **295**, 09033 (2024).
- F. H. O'Shea**, S. Joung, D. R. Smith, R. Coffee, *Automatic identification of edge localized modes in the DIII-D tokamak*, APL Mach. Learn. **1**, 026102 (2023).
- R. Humble, **F. H. O'Shea**, W. Colocho, M. Gibbs, H. Chaffee, E. Darve, and D. Ratner, *Beam-based rf station fault identification at the SLAC Linac Coherent Light Source*, Phys. Rev. Accel. Beams **25** 122804 (2022).
- F. H. O'Shea**, N. Bruchon, and G. Gaio, *Policy gradient methods for free-electron laser and terahertz source optimization and stabilization at the FERMI free-electron laser at Elettra*, Phys. Rev - Accel. Beams **23**, 122802 (2020).
- N. Bruchon, G. Fenu, G. Gaio, M. Lonza, **F. H. O'Shea**, F. A. Pellegrino, E. Salvato, *Basic reinforcement learning techniques to control the intensity of a seeded free-electron laser*, Electronics **9**, 781 (2020).